Python MySQL Database Connectivity

Step 1: Install Python

Verify Python installation.

```
python --version
```

Output

```
Python 3.14.x
```

Step 2: Install MySQL

Verify MySQL installation.

```
SELECT VERSION();
```

Step 3: Install MySQL Connector

- Open Command Prompt and run

```
pip install mysql-connector-python
```

- Verify installation

```
pip show mysql-connector-python
```

Step 4: Create Database

```
CREATE DATABASE PythonDB;
```

Use the database

```
USE PythonDB;
```

Step 5: Create Table

```
CREATE TABLE Student
(
   id INT PRIMARY KEY AUTO_INCREMENT,
   name VARCHAR(100),
   age INT,
   city VARCHAR(50)
);
```

Step 6: Create Python File

Create

insert.py

Step 7: Import Library

```
import mysql.connector
```

Step 8: Create Connection

Replace your username and password.

```
import mysql.connector


connection = mysql.connector.connect(
    host="localhost",
    user="root",
    password="your_password",
    database="PythonDB"
)


print("Connected Successfully")
```

Step 9: Create Cursor

```
cursor = connection.cursor()
```

Step 10: Write Insert Query

```
sql = """
INSERT INTO Student(name, age, city)
VALUES(%s, %s, %s)
"""
```

Notice **%s** is used as the placeholder regardless of the data type.

Step 11: Store Values

```
values = ("Anuj", 25, "Mumbai")
```

Step 12: Execute Query

```
cursor.execute(sql, values)
```

Step 13: Save Changes

```
connection.commit()
```

Step 14: Print Result

```
print(cursor.rowcount, "Record Inserted")
```

Step 15: Close Connection

```
cursor.close()

connection.close()
```

Complete Program

```
import mysql.connector

connection = mysql.connector.connect(
    host="localhost",
    user="root",
    password="your_password",
    database="PythonDB"
)

cursor = connection.cursor()

sql = """
INSERT INTO Student(name, age, city)
VALUES(%s, %s, %s)
"""

values = ("Anuj", 25, "Mumbai")

cursor.execute(sql, values)

connection.commit()

print(cursor.rowcount, "Record Inserted Successfully")

cursor.close()
connection.close()
```

Output

Connected Successfully

1 Record Inserted Successfully

Inserting Multiple Records

```
import mysql.connector

connection = mysql.connector.connect(
    host="localhost",
    user="root",
    password="your_password",
    database="PythonDB"
)

cursor = connection.cursor()

sql = """
INSERT INTO Student(name, age, city)
VALUES(%s, %s, %s)
"""

students = [
    ("Rahul", 22, "Delhi"),
    ("Sneha", 24, "Pune"),
    ("Amit", 23, "Jaipur"),
    ("Priya", 21, "Nagpur")
]

cursor.executemany(sql, students)

connection.commit()

print(cursor.rowcount, "Records Inserted")

cursor.close()
connection.close()
```

Output

4 Records Inserted

Common Errors

Error	Cause	Solution
Access denied	Wrong username/password	Verify MySQL credentials
Unknown database	Database not created	Create the database first
Table doesn't exist	Wrong table name	Check the table name

ModuleNotFoundError: No module named 'mysql'	Connector not installed	Run pip install mysql-connector-python
Can't connect to MySQL server	MySQL service is not running	Start the MySQL server/service

Best Practices

- Use parameterized queries (%s) to prevent SQL injection.
- Always call connection.commit() after INSERT, UPDATE, or DELETE operations.
- Close the cursor and connection after use, or use a try...finally block (or context managers where supported) to ensure resources are released even if an error occurs.
- Never hard-code database passwords in production applications; use environment variables or a secure configuration file.